HU COLLECTION · KEVINVARIANT

Binomial Coefficients in Olympiad Problems

Kevin Hu
November 2026

Handout Combinatorics · Number theory

Section 1 fixes the definition. Section 2 proves the identities and the divisibility theorems that olympiad problems use, in an order where each result uses only earlier ones. Section 3 has nine real problems, from BMO1 level to IMO Problem 6, and Section 4 has their solutions.
Prerequisites. Induction, modular arithmetic, complex roots of unity. **Time.** About four hours, including the problems.

1 The definition

For integers $n \geq 0$ and k, the *binomial coefficient* $\binom{n}{k}$ is the number of k-element subsets of an n-element set. In particular $\binom{n}{0} = \binom{n}{n} = 1$, and $\binom{n}{k} = 0$ when $k < 0$ or $k > n$. This convention is worth keeping. Every sum below can then run over all integers k, and Pascal's rule needs no exceptions at the edges.

Theorem 1.1 (Factorial formula). For $0 \leq k \leq n$,

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!} = \frac{n(n-1)\cdots(n-k+1)}{k!}.$$

Proof. There are $n(n-1)\cdots(n-k+1)$ ordered lists of k distinct elements, and each k-subset gives exactly $k!$ of them. ■

Theorem 1.2 (Pascal's rule). For $n \geq 1$ and every integer k, $\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$.

Proof. Fix an element x. The k-subsets without x number $\binom{n-1}{k}$ and those with x number $\binom{n-1}{k-1}$. ■

Theorem 1.3 (Binomial theorem). $(x+y)^n = \sum_k \binom{n}{k} x^k y^{n-k}$.

Proof. Expand the product of n brackets $(x+y)$. The monomial $x^k y^{n-k}$ appears once for each choice of the k brackets that contribute x. ■

Writing row n as the list $\binom{n}{0}, \binom{n}{1}, \ldots, \binom{n}{n}$ and stacking the rows gives Pascal's triangle (Figure 1). Pascal's rule says that each entry is the sum of the two entries directly above.

© 2026 Kevin Hu · kevinvariant08.github.io · CC BY-NC-ND 4.0

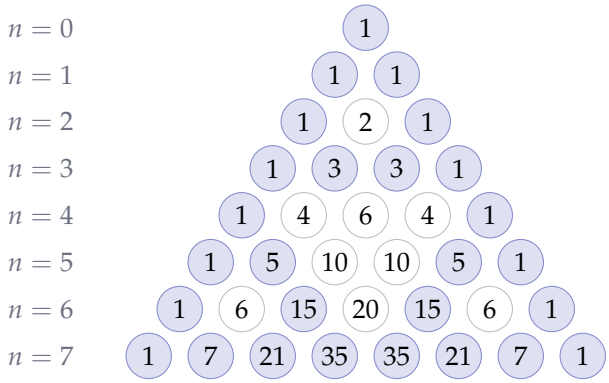


Figure 1: Rows 0 to 7 of Pascal's triangle. The odd entries are shaded. Section 2.3 explains the pattern.

2 Identities and theorems

2.1 Counting twice

Most identities in this subsection have a proof by counting one set in two ways. If one side is a sum over i, look for a set that splits into classes indexed by i.

Theorem 2.1 (Symmetry). $\binom{n}{k} = \binom{n}{n-k}$.

Proof. Taking complements is a bijection between k-subsets and $(n-k)$-subsets. ■

Theorem 2.2 (Row sums). $\sum_k \binom{n}{k} = 2^n$, and for $n \geq 1$, $\sum_k (-1)^k \binom{n}{k} = 0$. Hence for $n \geq 1$ the entries of row n with even k and those with odd k each sum to 2^{n-1}.

Proof. Put $(x, y) = (1, 1)$ and $(x, y) = (-1, 1)$ in Theorem 1.3. For a counting proof of the second identity, fix an element x and toggle x in or out of each subset. This pairs the subsets of even size with those of odd size. ■

Theorem 2.3 (Absorption). For $n \geq 1$ and every integer k, $k\binom{n}{k} = n\binom{n-1}{k-1}$.

Proof. Count pairs (S, x) with S a k-subset of an n-set and $x \in S$ (a committee with a chair). Choosing S first gives $k\binom{n}{k}$. Choosing the chair first and then the other $k-1$ members gives $n\binom{n-1}{k-1}$. ■

Absorption removes a factor k standing next to $\binom{n}{k}$. For example $\sum_k k\binom{n}{k} = n\sum_k \binom{n-1}{k-1} = n2^{n-1}$.

Theorem 2.4 (Hockey stick). For $0 \leq k \leq n$, $\sum_{m=k}^{n} \binom{m}{k} = \binom{n+1}{k+1}$.

Proof. Classify the $(k+1)$-subsets of $\{1, 2, \ldots, n+1\}$ by their largest element. If the largest element is $m+1$, the other k elements come from $\{1, \ldots, m\}$, in $\binom{m}{k}$ ways, and m ranges from k to n. ■

In the triangle the summands run down a diagonal and the total sits one step below and across from the last of them, which gives the identity its name.

Theorem 2.5 (Vandermonde's identity). For $m, n \geq 0$ and every k, $\displaystyle\sum_{i} \binom{m}{i}\binom{n}{k-i} = \binom{m+n}{k}$.

Proof. Let A and B be disjoint sets of sizes m and n. Classify the k-subsets S of $A \cup B$ by $i = |S \cap A|$. ∎

Corollary 2.6 (Sum of squares of a row). $\displaystyle\sum_{i} \binom{n}{i}^2 = \binom{2n}{n}$.

Proof. Take $m = n = k$ in Theorem 2.5 and use $\binom{n}{n-i} = \binom{n}{i}$. ∎

Theorem 2.7 (Shallow diagonals). Let $F_1 = F_2 = 1$ and $F_{m+2} = F_{m+1} + F_m$. Then for every $n \geq 0$,

$$\sum_{i \geq 0} \binom{n-i}{i} = F_{n+1}.$$

Proof. Let t_n be the number of tilings of a $1 \times n$ strip by 1×1 squares and 1×2 dominoes. Looking at the last tile gives $t_n = t_{n-1} + t_{n-2}$, and $t_0 = t_1 = 1$, so $t_n = F_{n+1}$. A tiling with i dominoes has $n - 2i$ squares, hence $n - i$ tiles, and is determined by which i of these $n - i$ tiles are dominoes. So there are $\binom{n-i}{i}$ such tilings. ∎

The terms $\binom{n}{0}, \binom{n-1}{1}, \binom{n-2}{2}, \dots$ lie on a line less steep than the sides of the triangle, hence the name.

Theorem 2.8 (Roots of unity filter). Let $f(x) = \sum_k a_k x^k$ be a polynomial, $m \geq 1$ and $\omega = e^{2\pi i/m}$. Then for every integer r,

$$\sum_{k \equiv r \pmod m} a_k = \frac{1}{m} \sum_{j=0}^{m-1} \omega^{-jr} f(\omega^j).$$

Proof. The inner sum $\sum_{j=0}^{m-1} \omega^{j(k-r)}$ equals m when $m \mid k - r$. Otherwise the sum is a geometric series whose ratio $\omega^{k-r} \neq 1$ satisfies $(\omega^{k-r})^m = 1$, so the sum is 0. Multiply by a_k and sum over k. ∎

For $m = 3$ and $f(x) = (1 + x)^n$ we have $1 + \omega = e^{i\pi/3}$ and $1 + \omega^2 = e^{-i\pi/3}$, so

$$\sum_{k \equiv r \pmod 3} \binom{n}{k} = \frac{2^n + 2\cos\frac{(n-2r)\pi}{3}}{3}.$$

The cosine term is one of $\pm 1, \pm 2$, so the three sums differ from $2^n/3$ by less than 1.

2.2 Central coefficients and Catalan numbers

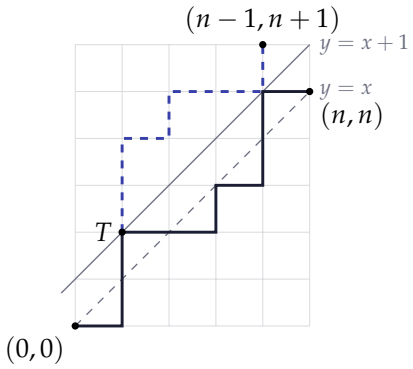
Theorem 2.9 (Central binomial coefficient). For $n \geq 0$, $\dfrac{4^n}{2n+1} \leq \binom{2n}{n} \leq 4^n$.

Proof. Since $\binom{2n}{k+1}/\binom{2n}{k} = \frac{2n-k}{k+1}$ is at least 1 exactly when $k < n$, the middle entry is the largest of the $2n+1$ entries of row $2n$. These entries sum to 4^n by Theorem 2.2. ■

Theorem 2.10 (Catalan numbers). The number of lattice paths from $(0,0)$ to (n,n) with steps $R = (1,0)$ and $U = (0,1)$ that never pass above the line $y = x$ is

$$C_n = \binom{2n}{n} - \binom{2n}{n+1} = \frac{1}{n+1}\binom{2n}{n}.$$

Proof. There are $\binom{2n}{n}$ paths in all. Call a path *bad* if the path passes above $y = x$. A bad path meets the line $y = x+1$, and we let T be the first point where this happens. Reflect the part of the path after T in the line $y = x+1$. This swaps the R and U steps after T and sends the end point (n,n) to $(n-1,n+1)$.
Conversely, a path from $(0,0)$ to $(n-1,n+1)$ starts below the line $y = x+1$ and ends above the line, so the path has a first point on the line, and reflecting after that point gives a bad path. The two maps are inverse to each other. So there are $\binom{2n}{n-1} = \binom{2n}{n+1}$ bad paths. Finally $\binom{2n}{n+1} = \frac{n}{n+1}\binom{2n}{n}$. ■



The first values are $C_0, \ldots, C_6 = 1, 1, 2, 5, 14, 42, 132$. The same numbers count balanced bracket sequences of length $2n$, triangulations of a convex $(n+2)$-gon and binary trees with n internal vertices. In a problem, a Catalan number usually appears through the recursion $C_{n+1} = \sum_{i=0}^{n} C_i C_{n-i}$, and Theorem 2.10 then supplies the closed form.

2.3 Binomial coefficients modulo a prime

Theorem 2.11. If p is prime and $0 < k < p$, then $p \mid \binom{p}{k}$.

Proof. $p! = k!\,(p-k)!\binom{p}{k}$. The prime p divides the left side but no factor of $k!\,(p-k)!$. ■

Corollary 2.12 (Freshman's dream). For polynomials with integer coefficients, $(1+x)^p \equiv 1 + x^p \pmod{p}$, meaning that corresponding coefficients are congruent. By induction, $(1+x)^{p^j} \equiv 1 + x^{p^j} \pmod{p}$ for all $j \geq 0$.

Proof. The first statement is Theorem 2.11. For the induction, $(1+x)^{p^{j+1}} = \left((1+x)^{p^j}\right)^p \equiv (1 + x^{p^j})^p \equiv 1 + x^{p^{j+1}}$, applying the first statement with x^{p^j} in place of x. ■

Theorem 2.13 (Lucas). Let p be prime and write $n = \sum_i n_i p^i$ and $k = \sum_i k_i p^i$ in base p, with digits $0 \leq n_i, k_i \leq p-1$. Then

$$\binom{n}{k} \equiv \prod_i \binom{n_i}{k_i} \pmod{p}.$$

Proof. By Corollary 2.12,

$$(1+x)^n = \prod_i \left((1+x)^{p^i}\right)^{n_i} \equiv \prod_i (1+x^{p^i})^{n_i} = \prod_i \sum_{j=0}^{n_i} \binom{n_i}{j} x^{jp^i} \pmod p.$$

Choosing a term j_i from the ith factor gives the monomial $x^{\sum j_i p^i}$ with $0 \le j_i \le n_i \le p-1$. By uniqueness of base p expansions, the exponent k arises only from $j_i = k_i$ for every i, and only if $k_i \le n_i$ for every i. So the coefficient of x^k on the right is $\prod_i \binom{n_i}{k_i}$, which is 0 when some $k_i > n_i$. Compare with the coefficient $\binom{n}{k}$ on the left. ■

Corollary 2.14. The number of k with $p \nmid \binom{n}{k}$ is $\prod_i (n_i+1)$. In particular, the number of odd entries in row n is $2^{s(n)}$, where $s(n)$ is the number of 1s in the binary expansion of n.

Proof. For digits $0 \le k_i \le n_i < p$ the number $\binom{n_i}{k_i}$ is not divisible by p, since $p \nmid n_i!$. So by Lucas $p \nmid \binom{n}{k}$ if and only if $k_i \le n_i$ for all i, and there are $n_i + 1$ choices for each digit k_i. ■

Figure 2 shows rows 0 to 31 modulo 2. If $2^m \le n < 2^{m+1}$, write $n = 2^m + n'$. By Corollary 2.12, $(1+x)^n \equiv (1+x^{2^m})(1+x)^{n'} \pmod 2$, so

$$\binom{n}{k} \equiv \binom{n'}{k} + \binom{n'}{k-2^m} \pmod 2.$$

Since $n' < 2^m$, at most one term on the right is nonzero. Hence rows 2^m to $2^{m+1}-1$ are two copies of rows 0 to $2^m - 1$ placed side by side, with even entries between them. The first 2^{m+1} rows are three copies of the first 2^m rows around an inverted triangle of even entries, which is the Sierpiński triangle, and by induction they contain exactly 3^{m+1} odd entries.

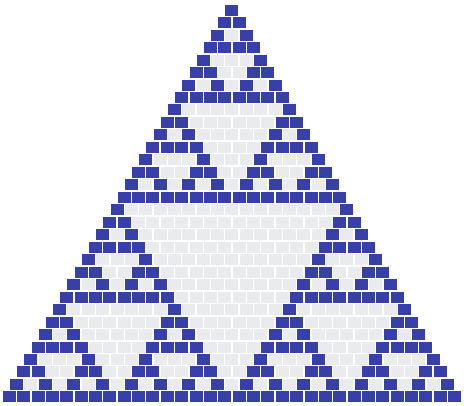


Figure 2: Rows 0 to 31 of Pascal's triangle modulo 2. Odd entries are dark.

For prime powers we work with $\mathbb{Z}_{(p)}$, the rationals whose denominator is not divisible by p. For $a, b \in \mathbb{Z}_{(p)}$ we write $a \equiv b \pmod{p^e}$ when $(a-b)/p^e \in \mathbb{Z}_{(p)}$. Sums and products of such congruences behave as for integers, every element of $\mathbb{Z}_{(p)}$ not divisible by p has an inverse in $\mathbb{Z}_{(p)}$, and for integers the new congruence agrees with the usual one. The next lemma is the form of Lucas's theorem that survives modulo p^e, for rows $n \equiv -1 \pmod{p^e}$.

Lemma 2.15 (Block lemma). Let p be prime, $e \ge 1$, and $p^e \mid n+1$. Put $n' = \frac{n+1}{p} - 1$. Then for $0 \le i \le n$,

$$\binom{n}{i} \equiv (-1)^{i-\lfloor i/p \rfloor} \binom{n'}{\lfloor i/p \rfloor} \pmod{p^e}.$$

Proof. Write $\binom{n}{i} = \prod_{j=1}^{i} \frac{n+1-j}{j}$. If $p \nmid j$ then $\frac{n+1-j}{j} = \frac{n+1}{j} - 1 \equiv -1 \pmod{p^e}$, because $p^e \mid n+1$ and j is invertible in $\mathbb{Z}_{(p)}$. If $j = pt$ then $\frac{n+1-pt}{pt} = \frac{n'+1-t}{t}$. With $q = \lfloor i/p \rfloor$, the multiples of p up to i are $p, 2p, \ldots, qp$, so

$$\binom{n}{i} = V \prod_{t=1}^{q} \frac{n'+1-t}{t} = V \binom{n'}{q}, \qquad V \equiv (-1)^{i-q} \pmod{p^e}.$$

Hence $\binom{n}{i} - (-1)^{i-q}\binom{n'}{q} = \binom{n'}{q}\left(V - (-1)^{i-q}\right)$ is an integer and is divisible by p^e in $\mathbb{Z}_{(p)}$, so also in $\mathbb{Z}$. ■

For $n = p-1$ the lemma reads $\binom{p-1}{i} \equiv (-1)^i \pmod{p}$, which is worth remembering on its own. The lemma says that row n modulo p^e is row n' with every entry repeated p times and with signs attached. Problems 6 and 9 use the lemma.

Theorem 2.16 (Wolstenholme). For every prime $p \geq 5$, $\binom{2p-1}{p-1} \equiv 1 \pmod{p^3}$, or equivalently $\binom{2p}{p} \equiv 2 \pmod{p^3}$.

Proof. The two forms are equivalent because $\binom{2p}{p} = 2\binom{2p-1}{p-1}$ by absorption. Now

$$\binom{2p-1}{p-1} = \prod_{j=1}^{p-1} \frac{p+j}{j} = \prod_{j=1}^{p-1} \left(1 + \frac{p}{j}\right) = 1 + pe_1 + p^2 e_2 + p^3 R,$$

where $e_1 = \sum_j \frac{1}{j}$, $e_2 = \sum_{i<j} \frac{1}{ij}$ and $R \in \mathbb{Z}_{(p)}$. We show $p^2 \mid e_1$ and $p \mid e_2$ in $\mathbb{Z}_{(p)}$.

Pair j with $p-j$. Then $2e_1 = \sum_{j=1}^{p-1} \left(\frac{1}{j} + \frac{1}{p-j}\right) = p\sum_{j=1}^{p-1} \frac{1}{j(p-j)}$, and $\frac{1}{j(p-j)} \equiv -\frac{1}{j^2} \pmod{p}$. As j runs over $1, \ldots, p-1$, so does j^{-1} modulo p, so

$$\sum_{j=1}^{p-1} \frac{1}{j^2} \equiv \sum_{j=1}^{p-1} j^2 = \frac{(p-1)p(2p-1)}{6} \equiv 0 \pmod{p},$$

using $p \geq 5$ to divide by 6. Hence $p^2 \mid e_1$. Finally $2e_2 = e_1^2 - \sum_j \frac{1}{j^2} \equiv 0 \pmod{p}$. ■

The theorem fails for $p = 3$, as $\binom{5}{2} = 10$. The pairing $j \leftrightarrow p-j$ comes back in Problem 8 with p replaced by a power of 2.

2.4 Exact powers of a prime

We write $v_p(x)$ for the exponent of p in a nonzero rational x, and $s_p(n)$ for the sum of the base p digits of n.

Theorem 2.17 (Legendre). $v_p(n!) = \sum_{i \geq 1} \left\lfloor \frac{n}{p^i} \right\rfloor = \frac{n - s_p(n)}{p-1}$.

Proof. $v_p(n!) = \sum_{m \leq n} v_p(m)$, and $v_p(m)$ is the number of $i \geq 1$ with $p^i \mid m$. Swapping the order of counting, each i contributes the $\lfloor n/p^i \rfloor$ multiples of p^i up to n. For the second form write $n = \sum_j n_j p^j$, so that $\lfloor n/p^i \rfloor = \sum_{j \geq i} n_j p^{j-i}$ and

$$\sum_{i \geq 1} \left\lfloor \frac{n}{p^i} \right\rfloor = \sum_j n_j (p^{j-1} + \cdots + p + 1) = \sum_j n_j \frac{p^j - 1}{p-1} = \frac{n - s_p(n)}{p-1}.$$ ■

Theorem 2.18 (Kummer). $v_p\binom{m+n}{m}$ equals the number of carries when m and n are added in base p.

Proof. By Legendre,

$$v_p\binom{m+n}{m} = \frac{s_p(m)+s_p(n)-s_p(m+n)}{p-1}.$$

Add m and n digit by digit. Each carry replaces a digit sum $d \geq p$ by $d-p$ and adds 1 to the next position, so the total digit sum drops by $p-1$. Without carries the digit sums simply add. So $s_p(m)+s_p(n)-s_p(m+n)$ is $p-1$ times the number of carries. ■

For example, adding n to itself in binary carries once at each digit 1, so $v_2\binom{2n}{n} = s_2(n)$. In particular $v_2\binom{2^k}{2^{k-1}} = 1$.

Theorem 2.19 (Prime factors of $\binom{2n}{n}$). Let $n \geq 1$ and let p be prime.

(a) If $p^a \mid \binom{2n}{n}$ then $p^a \leq 2n$. Hence $\binom{2n}{n}$ divides $\operatorname{lcm}(1,2,\ldots,2n)$.

(b) If $n < p \leq 2n$ then $v_p\binom{2n}{n} = 1$.

(c) If $p \geq 3$ and $\frac{2n}{3} < p \leq n$ then $p \nmid \binom{2n}{n}$.

Proof. By Legendre, $v_p\binom{2n}{n} = \sum_{i\geq 1}\left(\lfloor 2n/p^i\rfloor - 2\lfloor n/p^i\rfloor\right)$. Every term is 0 or 1, because $\lfloor 2x\rfloor - 2\lfloor x\rfloor \in \{0,1\}$ for real x, and the term vanishes once $p^i > 2n$. So $v_p\binom{2n}{n}$ is at most the number of $i \geq 1$ with $p^i \leq 2n$, which proves (a), since $v_p \operatorname{lcm}(1,\ldots,2n)$ is exactly that number. In (b) and (c) we have $p^2 > 2n$ (in (c) because $p^2 \geq 3p > 2n$), so only the term $i = 1$ survives. In (b) the term is $1-0=1$, and in (c) the term is $2-2\cdot 1 = 0$. ■

Remark. (Erdős's proof of Bertrand's postulate.) Suppose no prime lies in $(n, 2n]$. By (b) and (c) every prime factor of $\binom{2n}{n}$ is at most $2n/3$. By (a), primes $p > \sqrt{2n}$ divide $\binom{2n}{n}$ at most once, and each of the fewer than $\sqrt{2n}$ primes $p \leq \sqrt{2n}$ contributes at most $2n$. Together with the bound $\prod_{p\leq x} p < 4^x$ (an induction on x that uses $\binom{2m+1}{m} \leq 4^m$) and Theorem 2.9,

$$\frac{4^n}{2n+1} \leq \binom{2n}{n} \leq (2n)^{\sqrt{2n}}\, 4^{2n/3}.$$

This fails for every $n \geq 468$. For smaller n use the primes $2, 3, 5, 7, 13, 23, 43, 83, 163, 317, 631$, each less than twice the one before.

2.5 How to spot a binomial problem

1. A sum of $\binom{n}{k}$ times a polynomial in k. Absorb the factors of k (Theorem 2.3), or count pairs.
2. A sum of products $\binom{a}{i}\binom{b}{k-i}$, or of squares $\binom{n}{i}^2$. Vandermonde.
3. A sum $\sum_m \binom{m}{k}$ over the top index, or an average of the least element over all subsets. Hockey stick.
4. A sum over k in a residue class, or a count of subsets with sum divisible by m. Roots of unity filter.
5. A sum $\sum \binom{N}{2k+1}a^k$. This is the coefficient of $\sqrt{a}$ in $(1+\sqrt{a})^N$, so pair $(1+\sqrt{a})^N$ with $(1-\sqrt{a})^N$.
6. Divisibility by a prime p. Theorem 2.11 and Wolstenholme when the numbers are below $2p$, Lucas in general.

7. An exact power of p, or a ratio of factorials claimed to be an integer. Legendre and Kummer, one prime at a time.
8. A row n with $n \equiv -1 \pmod{p^e}$, typically $n = 2^k - 1$ or $n = p - 1$. The block lemma.

3 Problems

The problems are in rough order of difficulty, from ⋆ (BMO1) to ⋆⋆⋆⋆⋆ (IMO Problem 6). Try each one before reading Section 4.

Problem 1 (Putnam 2000, B2) ⋆ Prove that the expression

$$\frac{\gcd(m,n)}{n}\binom{n}{m}$$

is an integer for all pairs of integers $n \geq m \geq 1$.

Problem 2 (IMO 1981, Problem 2) ⋆ Let $1 \leq r \leq n$ and consider all subsets of r elements of the set $\{1, 2, \ldots, n\}$. Each of these subsets has a smallest member. Let $F(n,r)$ denote the arithmetic mean of these smallest numbers; prove that

$$F(n,r) = \frac{n+1}{r+1}.$$

Problem 3 (IMO 1972, Problem 3) ⋆⋆ Let m and n be arbitrary non-negative integers. Prove that

$$\frac{(2m)!\,(2n)!}{m!\,n!\,(m+n)!}$$

is an integer. ($0! = 1$.)

Problem 4 (IMO 1974, Problem 3) ⋆⋆ Prove that the number $\sum_{k=0}^{n} \binom{2n+1}{2k+1} 2^{3k}$ is not divisible by 5 for any integer $n \geq 0$.

Problem 5 (Putnam 1991, B4) ⋆⋆⋆ Suppose p is an odd prime. Prove that

$$\sum_{j=0}^{p} \binom{p}{j}\binom{p+j}{j} \equiv 2^p + 1 \pmod{p^2}.$$

Problem 6 (ISL 2008, N4) ⋆⋆⋆⋆ Let n be a positive integer. Show that the numbers

$$\binom{2^n-1}{0}, \binom{2^n-1}{1}, \binom{2^n-1}{2}, \ldots, \binom{2^n-1}{2^{n-1}-1}$$

are congruent modulo 2^n to $1, 3, 5, \ldots, 2^n - 1$ in some order.

Problem 7 (IMO 1995, Problem 6) ⋆⋆⋆⋆ Let p be an odd prime number. How many p-element subsets A of $\{1, 2, \ldots, 2p\}$ are there, the sum of whose elements is divisible by p?

Problem 8 (ISL 2007, N4) ⋆⋆⋆⋆⋆ For every integer $k \geq 2$, prove that 2^{3k} divides the number

$$\binom{2^{k+1}}{2^k} - \binom{2^k}{2^{k-1}}$$

but 2^{3k+1} does not.

© 2026 Kevin Hu · kevinvariant08.github.io · CC BY-NC-ND 4.0

Problem 9 (USAMO 2025, Problem 5) ⋆⋆⋆⋆⋆ Determine, with proof, all positive integers k such that

$$\frac{1}{n+1}\sum_{i=0}^{n}\binom{n}{i}^k$$

is an integer for every positive integer n.

4 Solutions

Problem 1 (Putnam 2000, B2)

Let $g=\gcd(m,n)$. By Bézout there are integers a,b with $g=am+bn$. Absorption (Theorem 2.3) gives $\frac{m}{n}\binom{n}{m}=\binom{n-1}{m-1}$, so

$$\frac{g}{n}\binom{n}{m}=a\cdot\frac{m}{n}\binom{n}{m}+b\binom{n}{m}=a\binom{n-1}{m-1}+b\binom{n}{m},$$

which is an integer. ■

Remark. A gcd next to a binomial coefficient is almost always handled this way. Write the gcd as a linear combination and treat each term separately.

Problem 2 (IMO 1981, Problem 2)

The number of r-subsets with smallest element k is $\binom{n-k}{r-1}$, and there are $\binom{n}{r}$ subsets in all. Writing $k=\sum_{j=1}^{k}1$ and swapping the order of summation, the sum of the smallest elements is

$$\sum_{k=1}^{n}k\binom{n-k}{r-1}=\sum_{j=1}^{n}\sum_{k=j}^{n}\binom{n-k}{r-1}=\sum_{j=1}^{n}\binom{n-j+1}{r}=\binom{n+1}{r+1},$$

using the hockey stick (Theorem 2.4) twice. Hence $F(n,r)=\binom{n+1}{r+1}/\binom{n}{r}=\frac{n+1}{r+1}$ by the factorial formula. ■

Remark. The double sum counts pairs (S,j) with $1\le j\le\min S$. The map $(S,j)\mapsto S\cup\{j-1\}$ is a bijection onto the $(r+1)$-subsets of $\{0,1,\dots,n\}$, with inverse "remove the least element t and put $j=t+1$". This gives the total $\binom{n+1}{r+1}$ with no computation.

Problem 3 (IMO 1972, Problem 3)

Fix a prime p and put $x_i=m/p^i$, $y_i=n/p^i$. By Legendre (Theorem 2.17), the exponent of p in the quotient is

$$\sum_{i\ge1}\Big(\lfloor2x_i\rfloor+\lfloor2y_i\rfloor-\lfloor x_i\rfloor-\lfloor y_i\rfloor-\lfloor x_i+y_i\rfloor\Big).$$

We claim that $\lfloor2x\rfloor+\lfloor2y\rfloor\ge\lfloor x\rfloor+\lfloor y\rfloor+\lfloor x+y\rfloor$ for all real x,y. Increasing x or y by 1 increases both sides by 2, so we may assume $0\le x,y<1$. The right side is then $\lfloor x+y\rfloor$. If $x+y<1$ the right side is 0. Otherwise the right side is 1 and $\max(x,y)\ge\frac12$, so the left side is at least 1. Hence every term of the sum is non-negative, the exponent of every prime is non-negative, and the quotient is an integer. ■

Remark. For $m=1$ the quotient is $2\binom{2n}{n}/(n+1)=2C_n$, twice a Catalan number. When a ratio of factorials has no visible counting meaning, compare the exponents of each prime as above.

Problem 4 (IMO 1974, Problem 3)

Let B be the given number and $A = \sum_{k=0}^{n} \binom{2n+1}{2k} 8^k$. By the binomial theorem,

$$(1+\sqrt{8})^{2n+1} = A + B\sqrt{8}, \qquad (1-\sqrt{8})^{2n+1} = A - B\sqrt{8},$$

and multiplying gives $A^2 - 8B^2 = (1-8)^{2n+1} = -7^{2n+1}$. Modulo 5 we have $8 \equiv 3$ and $-7^{2n+1} = -7 \cdot 49^n \equiv -2(-1)^n$, so

$$A^2 - 3B^2 \equiv \pm 2 \pmod 5.$$

If $5 \mid B$ then $A^2 \equiv \pm 2 \pmod 5$, which is impossible because the squares modulo 5 are $0, 1, 4$. ■

Remark. Read any sum $\sum_k \binom{N}{2k+1} a^k$ as the coefficient of $\sqrt{a}$ in $(1+\sqrt{a})^N$. The conjugate then gives a norm equation, here $A^2 - 8B^2 = -7^{2n+1}$, which is much easier to reduce modulo a prime than the sum itself.

Problem 5 (Putnam 1991, B4)

Let S be the sum. For $0 < j < p$ we have $p \mid \binom{p}{j}$ (Theorem 2.11) and $\binom{p+j}{j} \equiv 1 \pmod p$, because $j!\binom{p+j}{j} = (p+1)(p+2)\cdots(p+j) \equiv j! \pmod p$ and $p \nmid j!$. So $\binom{p}{j}\left(\binom{p+j}{j} - 1\right)$ is divisible by p^2, that is, $\binom{p}{j}\binom{p+j}{j} \equiv \binom{p}{j} \pmod{p^2}$. The terms $j = 0$ and $j = p$ are 1 and $\binom{2p}{p}$. Therefore

$$S \equiv 1 + \sum_{j=1}^{p-1} \binom{p}{j} + \binom{2p}{p} = 2^p - 1 + \binom{2p}{p} \pmod{p^2}.$$

By Corollary 2.6, $\binom{2p}{p} = \sum_j \binom{p}{j}^2 = 2 + \sum_{j=1}^{p-1} \binom{p}{j}^2 \equiv 2 \pmod{p^2}$, as each $\binom{p}{j}^2$ with $0 < j < p$ is divisible by p^2. Hence $S \equiv 2^p + 1 \pmod{p^2}$. ■

Problem 6 (ISL 2008, N4)

Let X_n be the set of residues modulo 2^n of the given numbers. There are 2^{n-1} odd residues modulo 2^n and 2^{n-1} given numbers, so we only need to prove that X_n is the set of all odd residues. We use induction on n. For $n = 1$ and $n = 2$ the numbers are 1, and $1, 3$. Let $n \geq 3$, assume the claim for $n-1$, and put $M = 2^{n-1}$, $c_q = \binom{M-1}{q}$ and $d_r = \binom{M/2-1}{r}$.

Step 1. The block lemma (Lemma 2.15) with $p = 2$, $e = n$ and row $2^n - 1$, for which $n' = M - 1$, gives

$$\binom{2^n-1}{2q} \equiv (-1)^q c_q, \qquad \binom{2^n-1}{2q+1} \equiv -(-1)^q c_q \pmod{2^n}.$$

As i runs from 0 to $M-1$, $q = \lfloor i/2 \rfloor$ runs from 0 to $M/2-1$, so $X_n = \{\pm c_q : 0 \leq q < M/2\}$ modulo 2^n. In the same way $X_{n-1} = \{\pm d_r : 0 \leq r < M/4\}$ modulo 2^{n-1}.

Step 2. We claim that, with $r = \lfloor q/2 \rfloor$,

$$(-1)^q c_q \equiv (-1)^r d_r + \lceil q/2 \rceil M \pmod{2^n}.$$

Indeed $(-1)^q c_q = \prod_{t=1}^{q} \frac{t-M}{t} = \prod_{t=1}^{q} \left(1 - \frac{M}{t}\right)$. For odd t we have $\frac{M}{t} - M = \frac{M(1-t)}{t}$ with $1-t$ even, so $1 - \frac{M}{t} \equiv 1 - M \equiv 1 + M \pmod{2^n}$. For even $t = 2t'$ the factor is $1 - \frac{M/2}{t'}$, and the product of these over $t' \leq r$ is $(-1)^r d_r$. There are $\lceil q/2 \rceil$ odd $t \leq q$, and $M^2 \equiv 0 \pmod{2^n}$, so

$$(-1)^q c_q \equiv (1+M)^{\lceil q/2 \rceil} (-1)^r d_r \equiv \left(1 + \lceil q/2 \rceil M\right)(-1)^r d_r \pmod{2^n}.$$

Since d_r is odd (Corollary 2.14, as $M/2-1$ has only 1s in binary), $Md_r \equiv M \pmod{2^n}$, and the claim follows.

Step 3. For each $r < M/4$, Step 2 with $q = 2r$ and $q = 2r+1$ gives $(-1)^r d_r + rM$ and $(-1)^r d_r + (r+1)M$. These are the two residues modulo 2^n that reduce to $(-1)^r d_r$ modulo 2^{n-1}. So the residues $(-1)^q c_q$ for $q < M/2$ are exactly the lifts to modulo 2^n of the residues $(-1)^r d_r$ modulo 2^{n-1} for $r < M/4$, and the same holds after negating both. By Step 1, X_n is the set of all lifts of X_{n-1}. By the induction hypothesis X_{n-1} is the set of odd residues modulo 2^{n-1}, so X_n is the set of odd residues modulo 2^n. ■

Remark. Step 1 alone loses information. Modulo 2^n the block lemma only refers back to row $M-1$ modulo 2^n, but the induction hypothesis knows that row only modulo 2^{n-1}. Step 2 runs the proof of the lemma one level further and keeps the error term $\lceil q/2 \rceil M$, which decides which lift occurs.

Problem 7 (IMO 1995, Problem 6)

The answer is $\dfrac{1}{p}\left(\binom{2p}{p} - 2\right) + 2$.

Let $\omega = e^{2\pi i/p}$ and $G(x,y) = \prod_{j=1}^{2p}(1+xy^j)$. The coefficient of $x^p y^s$ in G is the number of p-subsets of $\{1,\dots,2p\}$ with sum s. Applying the roots of unity filter (Theorem 2.8) in the variable y with $r = 0$, the required number is

$$N = \frac{1}{p}\sum_{s=0}^{p-1} [x^p]\, G(x, \omega^s),$$

where $[x^p]$ denotes the coefficient of x^p. For $s = 0$, $G(x,1) = (1+x)^{2p}$ contributes $\binom{2p}{p}$. For $1 \le s \le p-1$, as j runs from 1 to p the number ω^{sj} runs over all pth roots of unity ζ, since $p \nmid s$. Putting $z = -1/x$ in $\prod_\zeta (z-\zeta) = z^p - 1$ and multiplying by $(-x)^p$ gives

$$\prod_{\zeta}(1+x\zeta) = 1 - (-x)^p = 1 + x^p,$$

as p is odd. Since $\omega^{s(j+p)} = \omega^{sj}$, we get $G(x,\omega^s) = (1+x^p)^2$, whose coefficient of x^p is 2. Hence

$$N = \frac{\binom{2p}{p} + 2(p-1)}{p} = \frac{1}{p}\left(\binom{2p}{p} - 2\right) + 2. \qquad ■$$

Remark. There is also a counting proof. Apart from $\{1,\dots,p\}$ and $\{p+1,\dots,2p\}$, every p-subset A meets $\{1,\dots,p\}$ in a set B with $0 < |B| < p$. Adding 1 modulo p to the elements of B changes the sum of A by $|B| \not\equiv 0 \pmod{p}$. So these subsets fall into orbits of size p, each containing exactly one subset with sum divisible by p, and both exceptional subsets have sum divisible by p.

Problem 8 (ISL 2007, N4)

Let $N = 2^k$ and $D = \binom{2N}{N} - \binom{N}{N/2}$. Split $\binom{2N}{N} = \prod_{j=1}^{N} \frac{N+j}{j}$ by the parity of j. For $j = 2t$ the factor is $\frac{N/2+t}{t}$, and the product of these over $t \le N/2$ is $\binom{N}{N/2}$. Hence

$$\binom{2N}{N} = \binom{N}{N/2} P, \qquad P = \prod_{j \text{ odd}, j<N}\left(1 + \frac{N}{j}\right), \qquad D = \binom{N}{N/2}(P-1).$$

By Kummer, $v_2\left(\binom{N}{N/2}\right) = 1$, so we need $v_2(P-1) = 3k-1$.

Since $k \geq 2$, $N/2$ is even and the odd $j < N$ split into the 2^{k-2} pairs $\{j, N-j\}$ with $j < N/2$. For such a pair, $\frac{N}{j} + \frac{N}{N-j} = \frac{N^2}{j(N-j)}$, so

$$\left(1+\frac{N}{j}\right)\left(1+\frac{N}{N-j}\right) = 1+\frac{2N^2}{j(N-j)} = 1+2^{2k+1}u_j, \qquad u_j = \frac{1}{j(N-j)}.$$

Expanding $P = \prod_j (1+2^{2k+1}u_j)$ gives $P-1 = 2^{2k+1}U + R$ with $U = \sum_j u_j$, where every term of R has 2-adic valuation at least $4k+2$ because the u_j have odd denominators. As $4k+2 > 3k-1$, we only need $v_2(U) = k-2$.

Since $N - j \equiv -j \pmod{2^k}$, we have $u_j \equiv -j^{-2} \pmod{2^k}$. As j runs over the odd numbers below 2^{k-1}, so does j^{-1} modulo 2^{k-1}. So with $m = 2^{k-2}$,

$$U \equiv -\sum_j j^{-2} \equiv -\sum_{i=1}^{m}(2i-1)^2 = -\frac{m(2m-1)(2m+1)}{3} \pmod{2^{k-1}}.$$

The right side has 2-adic valuation exactly $k-2 < k-1$, so $v_2(U) = k-2$. Therefore $v_2(P-1) = (2k+1)+(k-2) = 3k-1$ and $v_2(D) = 3k$. ■

Remark. The pairing $j \leftrightarrow N-j$ is the step from the proof of Wolstenholme's theorem, with p replaced by 2^k. Pairing raises each error term from 2^k to 2^{2k+1}, so products of two error terms lie beyond the precision we need.

Problem 9 (USAMO 2025, Problem 5)

The answer is all even k. Let $S(n) = \sum_{i=0}^{n} \binom{n}{i}^k$.

If k is odd, take $n = 2$. Then $S(2) = 2 + 2^k \equiv 2 + 2 = 4 \equiv 1 \pmod 3$, so $3 \nmid S(2)$.

Let k be even. We prove $n+1 \mid S(n)$ for all $n \geq 0$ by strong induction, the case $n = 0$ being $S(0) = 1$. Let $n \geq 1$ and let p^e be the exact power of a prime p dividing $n+1$, with $e \geq 1$. Put $n' = \frac{n+1}{p} - 1 < n$. By the block lemma (Lemma 2.15), and since k is even,

$$\binom{n}{i}^k \equiv \binom{n'}{\lfloor i/p \rfloor}^k \pmod{p^e}.$$

As i runs from 0 to $n = p(n'+1)-1$, $\lfloor i/p \rfloor$ takes each value $0, 1, \ldots, n'$ exactly p times, so $S(n) \equiv pS(n') \pmod{p^e}$. By the induction hypothesis $n'+1 \mid S(n')$, and $p^{e-1} \mid \frac{n+1}{p} = n'+1$, so $p^e \mid pS(n')$ and hence $p^e \mid S(n)$. This holds for every prime power exactly dividing $n+1$, so $n+1 \mid S(n)$. ■

Remark. The block lemma turns a statement about row n modulo p^e into a statement about the shorter row n' modulo p^{e-1}, which is what strong induction needs. Try the lemma whenever a problem involves a whole row and the modulus divides $n+1$.

© 2026 Kevin Hu · kevinvariant08.github.io · CC BY-NC-ND 4.0